

**NATURAL SCIENCES
YEAR 4
TERM 1 TO 3
LEARNER'S BOOK**

CONTENT OVERVIEW

LIFE AND LIVING	<u>Living and non-living things</u> Living things • <u>Identify</u> and explain • Briefly explain the seven life processes • Non-living • Identify • Explain why things are non-living Practical: Living and Non-living things	<u>Digestive system</u> • Explain concepts • Describe structure • Name the parts and the organs • Describe the features and functions of the parts and organs	<u>Sense Organs</u> Senses • Briefly define concepts • Name and describe animal senses adapted for survival
	<u>Structure of plants</u> • Describe the basic structure of plants • Identify differences between plants	<u>Nutrients in food</u> Food Groups • Briefly define concepts • Categorise food • Describe the functions of food groups	<u>Human skeleton</u> • Briefly define concepts • Function of bones
	<u>Structure of animals</u> • Describe the basic structure of animals • Identify differences between animals	<u>Nutrition</u> • Explain a balanced diet	<u>Human Reproductive System</u> Reproduction • Introduce and explain concepts with examples • Explain the physical changes • Explain the features in the human organisms
	<u>What plants need to grow</u> • Explain requirements to growth • Explain how plants grow from cuttings and seeds	<u>Food processing</u> • Explain concepts, importance of preserving, types of products, methods and benefits	

CONTENT OUTLINE

WEEK	TOPIC	CONTENT
1-2	Sense Organs	Senses - Briefly define concepts through demonstration: - Senses - Sight(vision) - Smell - Hearing - Sense of touch - Taste - Name and describe animal senses adapted for survival: - Use hearing for protection etc. - Use smell to find food
3-5	Human skeleton	Human skeleton - Briefly define the basic concepts: - Muscles - The skull - The backbone - The ribs - Limbs - Shoulder - Hip girdles - Joints - Pelvis - Demonstrate and explain the number of bones that make up an adult human skeleton - Describe the function of the bones of the human skeleton - Explain how individual bones in the skeleton are held together - Discuss the importance of eating healthy foods to develop healthy and strong bones. - Discuss therapy and care for broken bones Practical: Draw a poster to illustrate and identify the physical changes in human development

6-8	Human Reproductive Reproduction	Human Reproductive System Reproduction • Introduce and explain concepts with examples - Sex hormones - Egg development - Sperm development - Menstrual cycle - Ovulation - Reproduction - Egg cells - Sperm cells - Reproductive system (males and females) • Explain and describe the prenatal development of a fetus during the three (3) trimesters in the uterus: • Explain the physical changes: - Explain the features in the human organisms development
9 - 10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 50 % • Test (Pen and Paper): 50 %		

YEAR 4 Term 1

TOPIC 1: SENSES

There are five senses the body has to react on external stimuli. It is important that learners define the new terminology.

Organ	Function
Eye	Sight/vision: using your eyes to see
Nose	Smell: Detection of odour and scents by using the nose.
Ear	Hearing: Detection of sound using your ears
Skin	Sense of touch: Bodily contact with an object
Tongue	Taste: The sensation of flavour on one's tongue when in contact with a substance

1.1. Senses

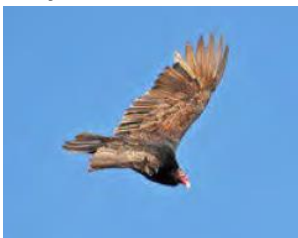
Animals can sense much more than humans and use their senses adapted for survival.

- a. Dogs for example have good sense of smell which they can use to determine the direction of an odour.



A dog

- b. Eagles, buzzards, hawks and other birds of prey have extremely sharp eyes as they have to see small rodents from very far away.



An eagle

- c. Elephants, cats and dogs can **hear** sounds that human **ears** cannot hear.

- d. Bats, dolphins and some whales use a special system called **echolocation**. Echolocation is a system animals use to determine the position of a prey by measuring how long it takes for an echo to return from the prey. They use this special sense to search for food and helps to find prey that is far away.



A dolphin

- e. Butterflies, bees and earthworms have another special sense called **chemoreceptors** - they taste through their skin or feet.



An earthworm

- f. Spiders are one of the animals that are sensitive to touch. They have a number of hairs on their legs. They are capable of detecting and tracing the origin of a vibration created by an object or another animal.

2. How animals use their senses to find food and for survival

Different animals use different senses to find food. Some animals use their sense of smell to protect them from other dangerous animals, e.g. skunks produce something that smells bad to the enemy to protect them; others use their sense smell to find their relatives and food. Other animals use more than one sense.

Finding Food	Protection
Birds of prey have excellent long-distance vision, but eagles stand out. The eagle has large eyes to see its prey from high in the sky.  <i>An eagle</i>	Some insects are usually bright in colours and warns the predators not to eat.

The owl has large eyes to find its food at night



An owl

Zebras have excellent eyesight, being able to see well during the day and at night. A combination of their good eyes, ears and noses increases their chances of sensing danger.



Zebra

Bees also use sight to get nectar.



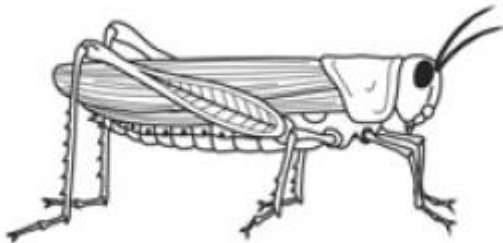
Bees about to collect nectar

A meerkat is always on the lookout for predators. Dark patches around their eyes reduce the glare of the sun so that they can be able to see predators clearly.



An alert meerkat

The thin antennae on the head of locust help them to smell and locate food.



The picture of a locust

In the African savannah zebras and the wildebeest tend to move around together. The wildebeest have a good sense of hearing and the zebra sees very well. In case of danger approaching, they can warn each other.



The picture showing zebras and wildebeest

ACTIVITY

Complete the following

1. Bats, whales and dolphins are able to use sound vibrations to help locate their food. What name is given to this special developed sense of hearing?
2. How does a skunk protect himself against predators?

By giving off a very terrible smell

3. Dogs have a highly developed sense of Smell that helps them to hunt.

The long, thin Antennae on the heads of ants help them to smell and locate food

ACTIVITY: Animal senses

Read the passage below and answer the questions that follow.

Humans have five senses; sight, hearing, taste, smell and touch. We use these senses every day of our lives when we work, play, eat and drink. Animals have senses too, and sometimes they are more powerful than ours.

But animals need to find their own food using their senses. For example, predators often have very good sight. Some birds of prey can see a mouse at a distance of 1.5 kilometres. However, they don't always catch the mouse. Mice have amazingly good hearing and this helps them escape from predators.

Insect sometimes have senses that are different from ours. For example, butterflies don't taste with their mouths –they taste with their feet.



A butterfly



A locust

Crickets use their legs to hear instead of their ears to communicate. Chameleons, like us, have two eyes, but they can move them independently and view in two distinct directions at once. Some animals appear to possess a unique sense that can predict the future.

A lot of species, including birds, ants, and many more, become more active when food is scarce. Because it's harder to find food in inclement weather, this aids in

their survival. They are aware of impending terrible weather. When flying a great distance, birds also employ this skill to avoid adverse weather.

Questions

1. How many senses do humans have?
 - five
2. What do animals use to find food?
 - They use their senses
3. What is unique about the eyes of a Chameleon?
 - They have two eyes that they can move separately and look in two different directions at the same time
4. What do crickets use to hear with?
 - They hear with their legs
5. How do butterflies taste?
 - Butterflies don't taste with their mouths but with their feet
6. Which sense do birds of prey use to get food?
 - They have very good sight.
7. Why is it sometimes difficult for birds of prey to catch mice?
 - Mice hear very well and this allow them to run away from predators
8. Which animals looks for food more busily before bad weather comes?
 - ants and birds

TOPIC 2: HUMAN SKELETON

KEY QUESTIONS

What does my skeleton look like?

Do all skeletons look like mine?

What are the functions of bones in my body?

How many bones does my body have?

What can I eat to keep my bones stronger?

Bones

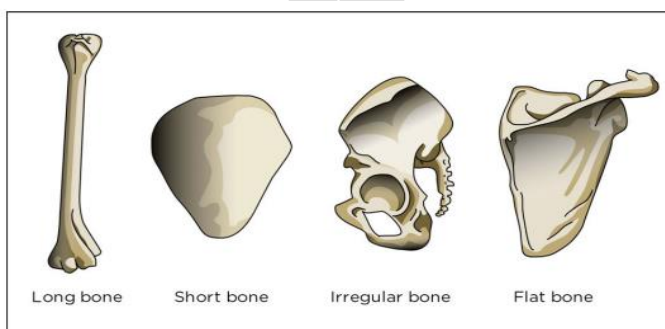
Bones are hard, and form a very strong frame structure to support and protect animal's body.

Bones that make up a human skeleton

The human skeleton is made up of 206 bones, including bones of the:

1. Skull – including the jaw bone
2. Spine – cervical, thoracic and lumbar vertebrae, sacrum and tailbone (coccyx)
3. Chest – ribs and breastbone (sternum)
4. Arms – shoulder blade (scapula), collar bone (clavicle), humerus, radius and ulna
5. Hands – wrist bones (carpals), metacarpals and phalanges
6. Pelvis – hip bones
7. Legs – thigh bone (femur), kneecap (patella), shin bone (tibia) and fibula (behind the shin bone)
8. Feet – tarsals, metatarsals and phalanges.

There are four different types of bone in the human body which are shown in the picture below.



ACTIVITY: Identifying bones in your body

MATERIALS

1. Photocopied bones puzzle picture of the human skeleton
2. Photocopied labels that go with the picture of the human skeleton
3. Scissors
4. Glue
5. Recycled thin cardboard such as a cereal box
6. Pencils and ruler
7. Colouring pencils if you want to decorate your skeleton

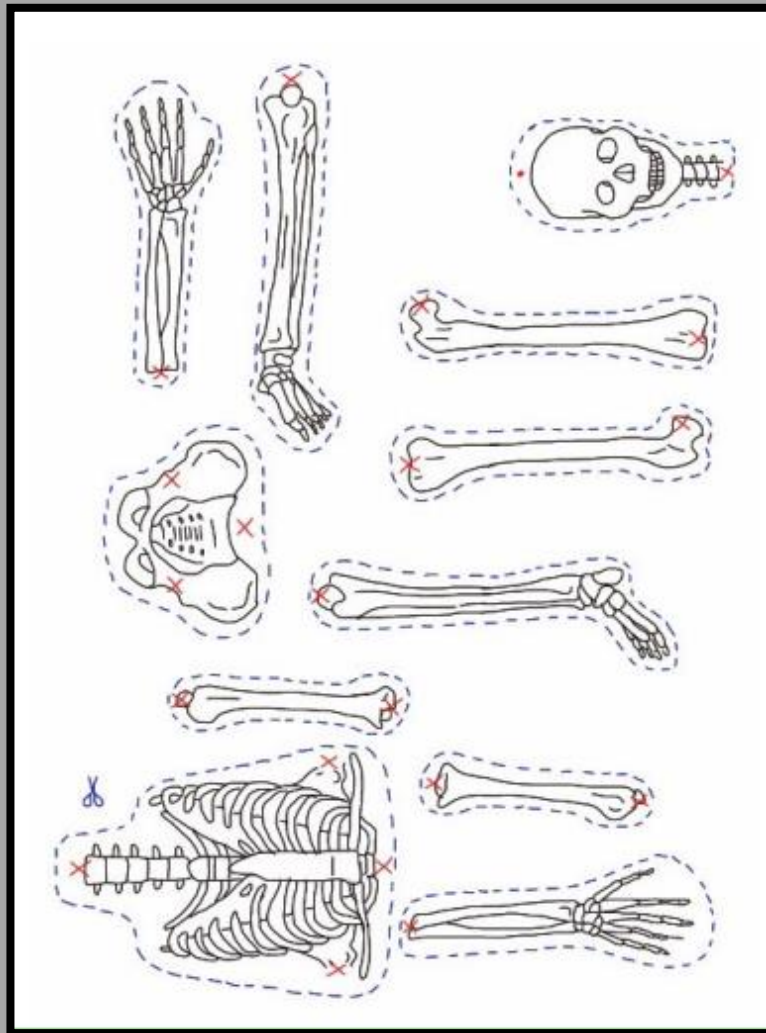
The labels below are for the bone puzzle and also need to be photocopied for the learners to cut out and add to their completed bones puzzles

skull	foot bones	lower jaw bone
thigh bone	ankle bones	inner forearm
calf bone	toe bones	shin bone
ribs	backbone	wrist bones
kneecaps	hip bone	finger bones
tail bone	collar bone	hand bones
breast bone	arm bone	outer forearm

Teachers can enlarge this puzzle and make a life-size version to hang or stick on the classroom door. Add labels and stick a small box to the door for suggestions for a name for the skeleton.

INSTRUCTIONS:

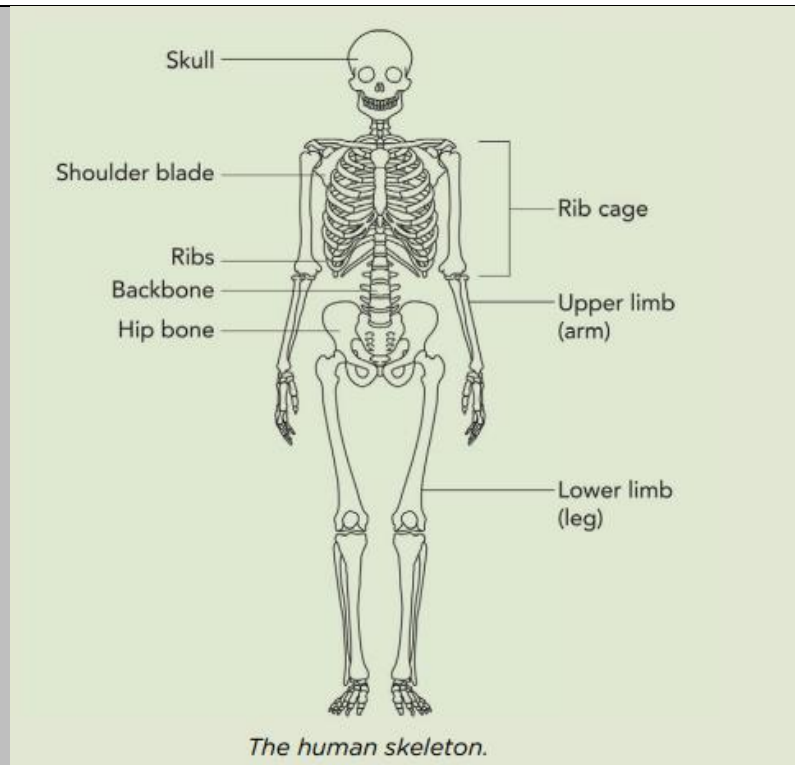
1. Your teacher will hand out a jumbled puzzle of the human skeleton. Carefully cut out each piece along the dotted line.
2. Build your human skeleton on the back of your recycled cardboard (cereal box) - do not stick it on yet as you might need to move it slightly if it does not fit properly onto the cardboard.
3. When you have it in place correctly, use glue to stick it to the cardboard.
4. Cut out the labels from the table.
5. Carefully pack the labels in the correct places - do not stick these down until you have done all of them as you might need to reposition them to fit it all in.



ACTIVITY: The bones in the human skeleton

INSTRUCTIONS:

1. Examine your skeleton puzzle. This illustration of the human skeleton might also help. Pay special attention to the shapes of different kinds of bones.
2. Identify examples of the four different kinds of bones? Write the examples of each kind of bone that you can find in this table.



Type of Bone	Where in the vertebrate body can you find it?
Long bones	
Short bones	
Flat bones	
Irregular bones	

The skull:

This is a collection of bones which encase the brain and give form to the head and face.

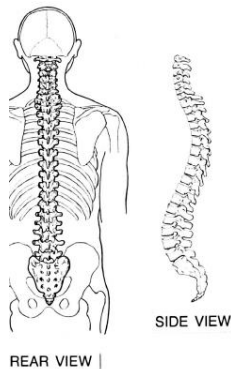
- The skull protects the eyes and ears, nose and mouth.
- It protects the brain.
- The teeth and the lower jaw is also attached to the skull.



The backbone:

The backbone is made up of vertebrae. The backbone has two functions:

- It protects the spinal cord that runs inside it.
- It supports the upper body.

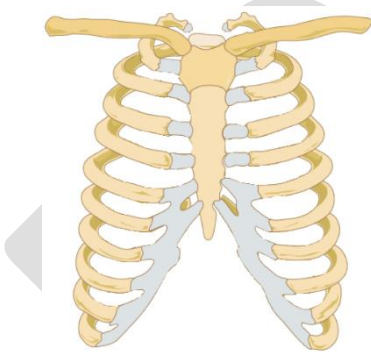


The ribs

Ribs are long curved bones around their chest.

The ribs are joined to the backbone and often to the front to form the rib cage.

In most vertebrates, the ribcage is around the chest area of the animal to protect the lungs, heart and other organs.



The shoulder:

This is the joint connecting the arm or forelimb with the body.



The *hip girdle*

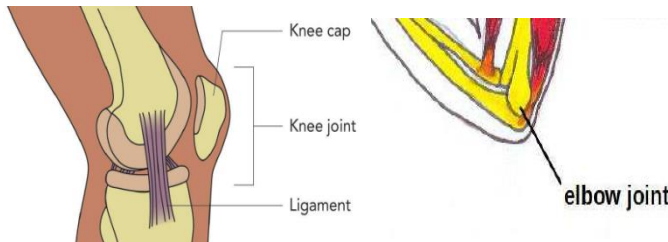
This is a ring-like bony structure, located in the lower part of the trunk.

It connects the axial skeleton to the lower limbs.



Joints

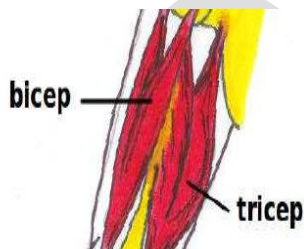
Joints are the places where bones come together. They come together in a special way to allow the animal or human to move - like at your elbow or wrist.



QUESTION

1. List four other joints in your skeleton that are not shown in the picture.

Muscles



These are tissues in the body that can tighten and relax to produce movement

Movement in human skeleton

Now that you know a little more about bones, let's see how humans move. Humans are able to move because skeleton has muscles, joints, tendons and ligaments.

Muscles are the largest of these four parts, but they would not be able to move the bones without joints, tendons, or ligaments. Muscles allow us to move because they are able to contract (become shorter) and relax (become longer).

Tendons connect muscle with bones. They work as levers to move your bones as your muscles contract and expand.

Joints open and close like a hinge (such as knees and elbows), whereas others allow for more complicated movement — a shoulder or hip joint, for example, allows for backward, forward,



The importance of eating healthy foods to develop strong bones.

A diet rich in calcium keeps your teeth and bones strong and can help to slow bone loss (osteoporosis) associated with getting older.

Calcium is usually associated with dairy products, but you can also get calcium by eating

- sardines, pilchards or tinned salmon (with bones)
- dark green vegetables - such as kale and broccoli
- calcium-fortified foods - such as soya products, fruit juices and cereals

THERAPY AND CARE FOR BROKEN BONES

ACTIVITY: Research and writing about therapy and care for broken bones.

INSTRUCTIONS:

1. You are going to learn about best ways in which to care for broken bones and how to help a person to recover from such an injury.
2. You will need to consult at least three different resources to find answers to the following questions:
 - What is the role of a doctor?
 - What is a fracture?
 - What are the signs and symptoms of a fracture?
 - What should one do if they break a bone?
 - How can we keep our bones healthy?
3. Present your findings on an A3 poster as part of an oral presentation (of 3 - 4 minutes) to the class.

Provide learners with the following information.

A fracture is a fancy word for a broken bone

There are numerous types of broken bones – they all differ based on the location and severity. Some can be simple fractures and some are complex.

Some heal with a splint and rest, some require and cast, and some might even require surgery

The signs and symptoms of broken bones include, pain, bruising, limping or “babying” a part of the body, deformity, swelling, tenderness.

When one breaks a bone.

They should keep the injury still (especially if neck or back injury), remain calm, tell an adult, who may splint the affected area, and go to the doctor.

An X-ray is a picture of your bones and provides a map for the doctor to see what direction to take with your injury.

Then, the doctor may cast the break so that the bone can heal or “knit” back together.

Casting is a big hard bandage.

Some breaks require a sling or crutches to help take pressure off the bone while it is healing.

How can we keep our bones healthy?

- Through exercise and good nutrition
- Assessment Rubric 1: Poster

Assessment criteria	0	1	2	Comments
Title	Absent	Present, but not sufficiently descriptive	Complete title	
Accuracy of facts	Many incorrect	Mostly correct, but some errors	All correct	
Language and spelling	Many errors	Some errors	No errors	
Organisation and layout	Disorganised and no logic	Organisation partially clear and logical	Excellent, logical layout	
Use of colour	No colour or only one colour	Some use of colour	Effective colour	
Accuracy of diagrams or pictures	Inaccurate	Mostly accurate	Completely accurate	
Creativeness	Nothing new or original	Some signs of creativity and independent thought	Original and very creative	
			Total	/14

TOPIC 3: HUMAN REPRODUCTIVE SYSTEM

1.1. Introduction

You have previously learnt that reproduction is one of the seven life processes, and like all organisms, humans need to reproduce to ensure the survival of the **species**. Humans reproduce sexually which means human reproduction requires a male and a female to mate to make an offspring (babies).

KEY QUESTIONS

What is reproduction?

What is reproductive system?

What is puberty and what does it mean when we "reach puberty"?

Why do we all go through puberty at different times and rates?

What changes take place inside our bodies during puberty?

What do our reproductive organs look like when they are mature?

How does a baby grow inside a woman's uterus?

Are there ways to prevent pregnancy and the transmission of STDs?

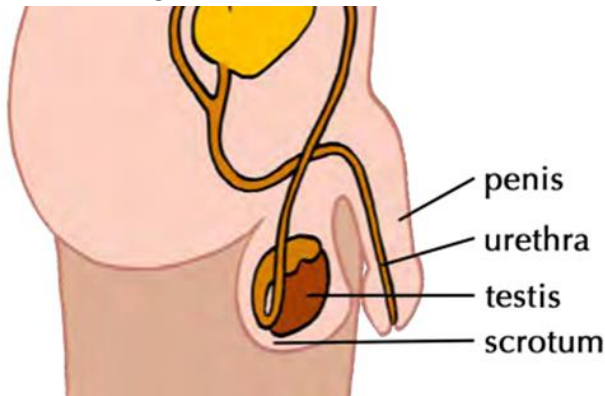
Reproduction is the process in which living things produce new offspring.

1.2. What is reproductive system?

Reproductive system is a system by which humans reproduce and bear a life offspring (babies). The main purpose of the reproductive system is to produce sex cells to ensure the continuation of the species.

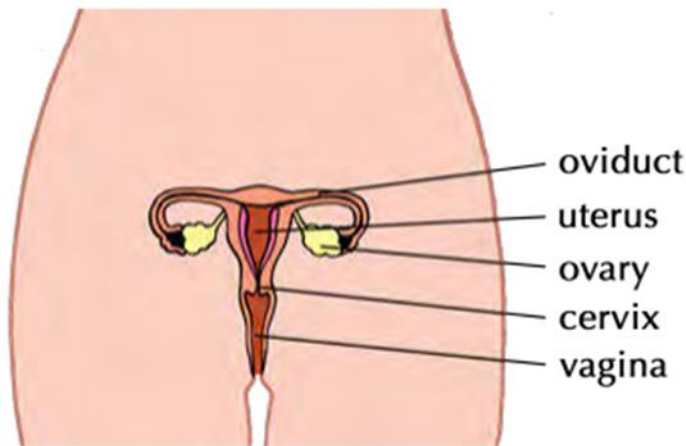
1.3. Parts of the reproductive system

2. In **males**, the reproductive organs include the penis and two **testes** hanging in a pouch or bag of skin called the **scrotum**.



The male reproductive organ

3. **The female reproductive organs** include the vagina, uterus, two fallopian tubes (oviducts) and two ovaries.



The female reproductive organ

3.2. Why do humans need to reproduce?

Humans need to reproduce in order for the Human species to survive.

3.3. Puberty

Puberty is the period in humans in which they experience physical changes in their bodies in order to be capable of sexual reproduction. The sexual organs mature to enable the body to produce sex cells. These sex cells are called **gametes**.

In human reproduction, two gametes (the sperm and egg cell) fuse during conception to form a zygote that will eventually become a new baby.

Our sexual organs are adapted for these functions and need to reach maturity. This

takes place during a stage in our life called puberty. Examples of female sex organs are vagina, uterus and ovaries. The examples of male sex organs are testicles and penis.

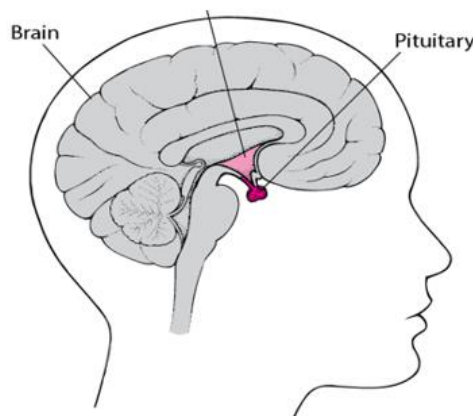
3.4. How does puberty start?

When a boy or girl reaches a certain point of growth and development, the sexual organs in the body also start to mature. Girls and boys do not, generally, go through puberty at exactly the same time:

- Girls go through puberty between 10/11 - 14/15 years of age
- Boys go through puberty between 12/13 - 15/16 years of age

During puberty, you will experience different physical and emotional changes as your body develops towards sexual maturity and adulthood. Many of the complex actions that take place in our bodies are controlled by chemical messengers called **hormones**.

Hormones are chemical substances that are produced from the pituitary gland (from the brain) and act like messenger molecules in the body.



The picture of the human brain

Puberty starts when the pituitary gland releases specific hormones that stimulate the testes and ovaries to release hormones **testosterone and oestrogen** that will start the production of **sperm** in the male and the maturation of ova (**egg cells**) in the female.

Sex hormones are chemical substances produced by a sex gland and has an effect on the sexual features of an organism.

Functions of testosterone

- It is the development of male secondary sexual characteristics, such as beard, pubic hair, deep voice and a muscular body.
- It stimulates the maturation of sperm cells.

Function of oestrogen

- It causes females to begin to menstruate, grow breasts and grow pubic and underarm hair, and may experience acne.
- It stimulates the ovaries to produce mature ova.

ACTIVITY: Let's take a look at some of the changes that take place during puberty.

INSTRUCTIONS:

1. Work in groups or pairs.
2. Discuss the changes that boys and girls go through during puberty and complete the table below.

QUESTIONS:

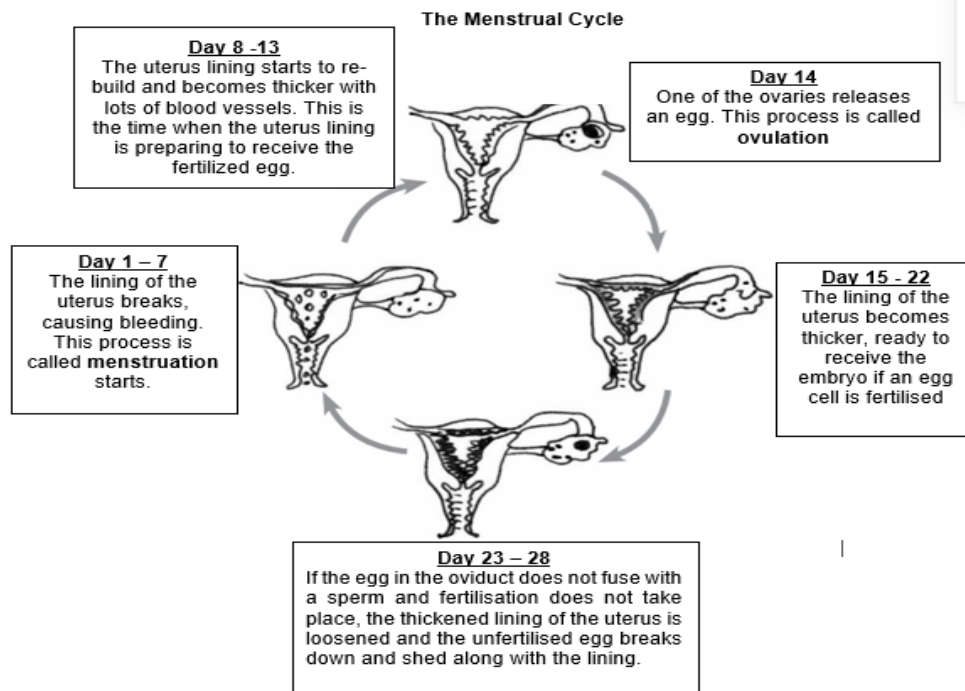
- Identify the changes that both go through during puberty and fill these into the table.

Changes in the girl	Changes in the boy

The following table summarises some of the physical changes that occur during puberty.

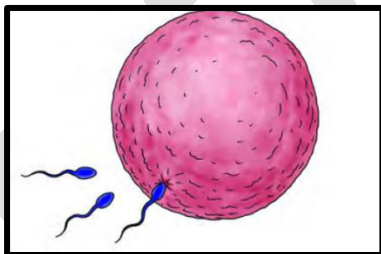
Physical changes in girls during puberty	Physical changes in boys during puberty
• The sexual organs (vagina, uterus and ovaries) start to mature.	• The sexual organs (testicles and penis) start to mature.
• Pubic hair starts to grow on the genitals.	• Pubic hair starts to grow on the genitals.

• Menstruation and fertility: girls have their first menstrual bleeding. • Menstruation is a sign that the ovaries have started secreting hormones and releasing eggs, and is therefore, a sign that the girl is now fertile (able to fall pregnant)	• Fertility: the testicles start to produce sperm that can fertilise a female egg during sexual intercourse.
• Body shape: changes occur, such as the waist becomes more defined and hips widen (to make space for childbirth). • There is an increase in body fat.	• Body shape: changes occur as the bones and muscle increase to give the young man a stronger, more muscular look.
• Breasts start to develop	• Voice changes occur becoming deeper and lower.
• Body odour is typically part of puberty as the skin produces more oils and the smell of sweat changes.	• Body odour is typically part of puberty as the skin produces more oils and the smell of sweat changes.
• Acne and pimples can occur due to changes in hormones and increased oil secretion from the skin.	• Acne and pimples can occur due to changes in hormones and increased oil secretion from the skin.



1.7 The prenatal development of a foetus during the three (3) trimesters in the uterus:

Pregnancy begins the moment the female egg cell is fertilised by the male sperm cell. This is called a **zygote**.

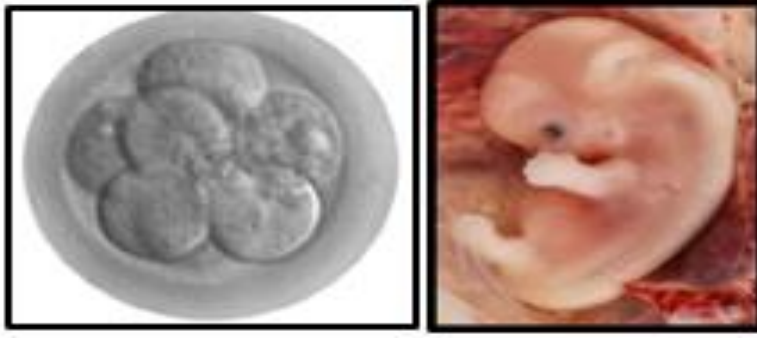


The picture showing the sperm cells fertilizing an ovum

Pregnancy is considered full term at 36 weeks. There are three trimester of pregnancy.

The First Trimester

The first trimester is the first 12 weeks of pregnancy. This is called the embryonic stage. The zygote divides itself and finally becomes an embryo. The baby's body begins to form. The placenta is partly formed by the mother and partly by the embryo. The embryo develops an umbilical cord to attach itself to the placenta. The embryo can receive food and oxygen and remove its wastes through the umbilical cord and placenta. The first organs to develop are the spinal cord, heart, brain. These will be followed by facial features eyes, ears, mouth, nasal passage, arms and legs.



The picture of an embryo

The Second Trimester

The second trimester of pregnancy begins at week 13-26. Once the embryonic stage ends the baby enters a foetal stage. At this stage the foetus begins sucking and swallowing, as well as moving inside the womb, although it may be too early to feel these movements.

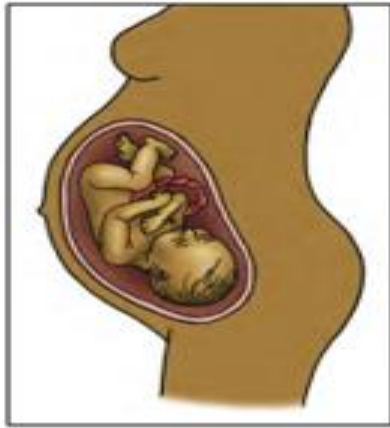
During the final weeks of the second trimester, fat starts to develop in the baby's skin. Organs such as the liver and pancreas become stronger.



The picture of a foetus

The Third Trimester

The third trimester of pregnancy begins at week 27 - 34. At this stage all the body's organs have developed and the movement inside the uterus increases. The baby gradually moves to a downward position.



Pregnancy in last trimester

ACTIVITY: Conduct a survey

INSTRUCTIONS:

1. Read some of the following comments people have made about menstruation and sex.
2. As a homework assignment, read these to at least five separate members of your family and friends.
3. Make a small cross if the person thinks it is not true and tick if they think it is true or correct.

Myth	Responses (tick or cross)
"You cannot fall pregnant during your period."	
"You cannot fall pregnant or make someone pregnant if you have sex in water."	
"Drinking and drugs make sex more fun."	
"Women are always moody and irrational during menstruation."	
." "If you have a shower after sex, you will not fall pregnant."	
"You cannot fall pregnant if it is your first time."	
"Everyone is having sex."	

QUESTIONS AND DISCUSSION

1. How many people in the homework survey believed that the comments were

in fact true?		
Most believed they were true	About half believed they were true and the other half believed they were not.	None believed they were true

2. Discuss with your class which of the comments were most widely believed to be true by the people you surveyed. Take some notes on the following lines.

3. Which of these comments had the strongest reaction from the people you surveyed? (Either positive or negative reaction.) Discuss their reactions with your class. Take some notes on the following lines.

4. Did anyone laugh at any of the comments? Which ones? Did they tell you why they laughed? Share this with the class.

5. Which of these statements do you think are true? Discuss this with your class and take some notes on the following lines.

Prevention of pregnancy and contraceptives

To avoid falling pregnant, you can use contraceptives. There are different contraceptives that can be used to prevent pregnancy. There are different types of contraceptives.

Male condoms are rubbery sheaths that are placed over a man's erect penis before sex, and are worn during sexual intercourse to prevent the sperm from entering the woman's vagina. Condoms also help to prevent the spread of STDs.

Female condoms also prevent the sperm from reaching the egg. However, these are placed in a woman's vagina to block the sperm, and are much less commonly used.

Oral contraceptive pills are used by many women today.

Many of these prevent ovulation. Pills need to be taken at the same time every day, otherwise they are not effective. While the pill is excellent at preventing pregnancy, it offers NO protection against STDs.

Sexually Transmitted Diseases (STDs)

There are dangerous and harmful diseases that are spread from one person to another during sexual intercourse and they are called Sexually Transmitted Diseases. STDs, for example are HIV/AIDS, Herpes virus, Syphilis, Gonorrhoea and genital warts. Some of these are life-threatening while others cause very painful and long-term symptoms.

You can prevent yourself from being infected with an STD by doing the following:

- **Get the facts:** Make sure you know up-to-date information about STDs and how they are spread, their symptoms and how you can protect yourself during sexual intercourse.
- **Take control of your sex life:** The more sexual partners you have, the higher your risk will be of having an STD.
- **Be faithful:** If you or your partner has sex with someone else, you are a risk of infecting the other person with the STD.
- **Using condoms** significantly reduces the risk of contracting STDs.

If you know that your partner has an STD, he or she can either get medical treatment for this and/or you can take the necessary precautions to prevent contracting the disease.

One of the most popular precautions to prevent the transmission of STDs is for the male partner to wear a condom.

It is your decision whether you want to participate in sexual intercourse with a romantic partner or not. There are two very important points to remember here:

- No one, no matter who they are, has the right to force you or pressure you to have sex with them or with anyone else. Therefore, you are the only one

who should be permitted to decide when you are ready to have a sexual relationship.

- If you decide to have sex, you should do so in a responsible manner. This includes protecting yourself against possible pregnancy and against any STD infection.

Revision

1. The reproductive organs are structured in a very specific way - to make fertilisation and pregnancy possible. Explain the function of each of these structures in the male and female bodies.

Reproductive organs	Their function
Ovaries	
Oviducts	
Uterus	
Testes	

2. A learner from your grade was trying to explain the process of the human reproduction cycle, but they scrambled up the order of the cycle. Write numbers 1 - 6 to place their sentences below into the correct order.

- _____ The sperm arrive in the oviduct.
- _____ During sexual intercourse, the sperm is propelled from the penis.
- _____ One sperm enters the outer cover of the egg to fertilise it.
- _____ The fertilised egg is implanted in the uterine lining.
- _____ The sperm travel from the vagina, through the uterus to the oviduct.
- _____ The egg is released from the ovaries and travels along the oviduct.

3. Define the following terms showing your own understanding puberty, menstruation, ovulation.

4. **Puberty.**

5. **Menstruation.**

6. **Ovulation**

TERM 2: MATTER AND MATERIALS

CONTENT OVERVIEW

MATTER AND MATERIALS	<u>States of matter</u> Solids, liquids and gases - Briefly define and describe three states - Distinguish between states - Describe the properties	<u>Mixtures</u> - Briefly define concepts - Distinguish between pure substances and mixtures - Separating of mixtures (demonstrate)	<u>Acids and bases</u> - Briefly define concepts - Describe difference between acids and bases - Test various household substances with blue and red litmus paper and sort them into acids, bases and neutrals Practical activity
	<u>Metals and non-metals</u> - Define concepts - Distinguish between metals and non-metals - Properties of metals and non-metals - Uses of metals and non-metals	<u>Solutions as special mixtures</u> - Ingredients of a mixture - Explain the difference between a mixture and solution - Describe the properties of a mixture	<u>Process to purify water</u> Clean Water - Introduce and explain concepts with examples - Importance of drinking clean water - Methods of cleaning water Practical Investigation
		<u>Dissolving</u> - Explain concept - Rate of dissolving Practical investigation: Rate of dissolving	<u>Impact of weather, temperature and wind on matter and materials.</u> - Impact of weather on materials - Ways to protect materials
	<u>Processing material</u> Briefly define	<u>The water cycle</u> Briefly define concepts	

	concepts • Explain processes to make new materials or products • List the properties of raw materials used to make new products Practical: Demonstrate the changes of raw material	• Describe the stages of the water cycle • Draw the water cycle and describe in your own words Practical	
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CONTENT OUTLINE

WEEK	TOPIC	CONTENT
1-3	Acids and bases	Acids and bases - Briefly define concepts: - Acids - Bases - Indicators - Universal indicator - Neutral - Neutralise - Group different acids and bases from household substances - Describe difference between acids and bases - Explain safe handling methods and disposal of household chemicals Practical activity: Test various household substances with blue and red litmus paper and sort them into acids bases and neutrals
4-6	Process to purify water	Process to purify water Clean Water - Introduce and explain concepts with examples: - Natural processes - Pollution - Filtering - Purify - Germ and bacteria - Explain the importance of drinking clean water - Demonstrate the different methods of cleaning water e.g. Processes as filtering, boiling, sieving etc. Practical Investigation: Design, draw or make your own water filter
7-8	Impact of weather on matter and materials	Impact of weather, temperature and wind on materials. - Impact of weather on materials - Metals - Non – metals

		• Discuss methods to protect materials against the weather
9-10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 40 % • Test (Pen and Paper): 60 %		

DRAFT

TOPIC 1: ACIDS AND BASES

KEY QUESTIONS

1. Which household substances contain acids and bases?
 2. How can we test if a substance is an acid, a base or a neutral substance?
- In this topic we are going to look at acids and bases.
 - Grouping different acids and bases from household substances.
 - Describing the differences between acids and bases and also looking at methods of handling and disposing of household chemicals.

1.1.1. Acid

The word acid comes from the Latin word '**acidus**', meaning 'sour'.

Acids taste sour, changes blue litmus paper to red, we have acids that are used at home and acids that are used in the laboratory.

Examples: Orange juice, lemon juice (citric acid), vinegar (acetic acid).

Acids used in laboratory are: hydrochloric acid, Nitric acid, sulphuric acid etc.

ACTIVITY

Below are statements. You need to state whether they are **True** or **False**.

1. Acids taste sour.
2. If we want to know if something is an acid, we can just taste it.
3. All acids are dangerous.
4. Vinegar is a mixture of small amount of acetic acid dissolved in water.
5. Laboratory acids must be handled with care and using protective clothing.
6. Formic acid is commonly referred to as vitamins.

1.1.2. Bases/ alkali

Bases taste bitter, feels slippery and turns red litmus blue. Soluble bases are called alkalis

Examples are: Bicarbonate of soda (baking soda) Tooth paste, bleach, oven cleaner, soap, ammonia,

Grouping household substances into acids and bases.

Let us group different household substances into acids and bases.

ACTIVITY

Acids and bases in our homes.

MATERIALS

- five bowls
- water
- washing powder
- handy Andy
- bicarbonate of soda or baking powder
- lemon juice
- vinegar
- a towel for drying hands
- Clean water bowl for washing hands after feeling each substance.

INSTRUCTIONS:

1. Place the bowls in a row on a desk in the front.
2. Mix each of the bases (washing powder, handy Andy, baking powder) in a separate bowls with some water.
3. Pour some lemon juice into another bowl and some vinegar into the last bowl.
4. Arrange the bowls in a row, alternating between a base and an acid.
Label the bowls
5. Put your finger in each of the bowls and feel the different substances.
6. Record what you feel and you should wash and dry your hands between each substance.
7. Copy the table below and record your observations.

Substance	How did it feel between the fingers?	Colour in blue litmus paper	Is it an acid or a base?
Washing powder			
Lemon juice			
Handy Andy/soup			
Vinegar			
Baking powder			

CONCLUSION

1. What can you conclude about how bases feel?
2. What can you conclude about how acids feel?
3. What did your teacher have to do to the dry washing powder before you could feel it in the bowl?
4. What do we call the solutions which forms?

1.1.3. Indicators

Indicators **show us whether something is a base or an acid. Indicators change colour when brought in contact with an acid or a base.** The most common indicators used in the laboratory is the litmus and universal indicator.

Some other substances also change colour when an acid or a base is added to them. By changing their colour, they show that they have reacted with an acid or a base. That is why we call them acid-base indicators. The most well-known acid-base indicator is a substance called **litmus**. It does not change colour in the presence of a neutral substance, but responds to acids and bases in the following way:

- litmus is red in the presence of an acid; and
- litmus is blue in the presence of a base.



Blue and red litmus paper.

1.1.4. Universal Indicator

A mixture of dyes that change colour to show accurately how acidic or basic a solution is.



Universal indicator paper

1.1.5. Neutral

Neutral is not an acid or a base .example: distilled water.

1.1.6. Neutralize

Adding equal amount of acids and bases in a solution to make it neutral.

DRAFT

INVESTIGATION: HOW DOES LITMUS RESPOND TO ACIDS AND BASES?

Aim: To determine how litmus responds to some household acids and bases.

MATERIALS AND APPARATUS

Small containers (test tubes or yoghurt tubs) filled with the following substances:

- Water
- Soda water
- Vinegar
- Lemon juice
- Baking soda (1 tablespoon dissolved in a cup of water)
- Handy Andy (1 tablespoon dissolved in a cup of water)
- Dishwashing liquid (1 teaspoon dissolved in a cup of water)
- Sugar solution
- Salt solution

(

- Litmus paper (blue and red)
- Glass or plastic rods (plastic teaspoons will also work well).

Method

1. Cut a small piece (1 cm long) of blue and red litmus for each substance that you will be testing.
2. Use the plastic teaspoon or rod to place just 1 drop of water on the blue litmus. Do the same with a piece of red litmus.
3. Did the blue litmus change colour? Did the red litmus change colour? Write the new colours in your table, in the appropriate places.
4. Repeat the procedure to test all the substances you have been given. You must rinse the teaspoon or rod with water in between substances.

Results and observations

Copy the table below and record your observations. If you did not use some of these substances, cross them out and write headings for your substances in the empty rows.

Substance	Prediction(acid or base)	Colour with blue litmus	Colour with red litmus	Results
Water				
Soda water				
Vinegar				
Lemon juice				
Baking soda				
Handy Andy				
Dishwashing Liquid				

Analysis

Let us now have a look at our observations to see what we can conclude.

1. How does the litmus paper indicate when a substance is an acid?
2. Which of the substances you tested are acids?
3. How does the litmus paper indicate when a substance is a base?
4. Which of the substances you tested are bases?
5. How would you describe a neutral substance?
6. How does the litmus paper indicate when a substance is neutral?
7. Which of the substances you tested were neutral?
8. Why do you think you had to rinse the glass rod or teaspoon in between testing each substance?

1.2. Differences between acids and bases

Acids	Bases
Acids turn blue litmus paper into red	Bases turn red litmus paper into blue.
Acid taste sour	Bases taste bitter
Rough	slippery

1.3. Safe handling methods and disposal of household chemicals.

Throwing chemicals items in your rubbish bin is dangerous. It can explode, ignite, leak or mix with other chemicals.

Pouring them down the drain will pollute the environment, harm animals and vegetation, contaminate our water supply and make rivers and beaches unsafe for swimming.

Follow these tips to safely store, use and dispose of toxic items at home.

Handling and storing chemicals

- Never mix chemicals during use or storage, as they may react.
- Handle chemicals carefully and avoid direct contact. Wear gloves and protective clothing where possible.
- Wash your hands after handling chemicals.

Disposing of chemicals

- Never dispose chemicals in your rubbish bin or drain.
- If you're unable to immediately dispose of household chemicals, make sure you safely store them until you can.

SUMMARY

Key concepts

1. Acids and bases are chemical opposites of each other.
2. Though it is not a good idea to taste chemicals, acids have a sour taste and bases taste bitter.
3. When they are in solution with water, acids feel rough and bases feel slippery.
4. Some acids and bases are present in foods and in household items. These are relatively safe to handle. Others are often very corrosive and should only be handled when you are wearing protective clothing.
5. Substances which are neither acidic nor basic, are called neutral substances.
6. When an acid is mixed with a base in the right quantities, they neutralise each other, which means they lose their power.
7. Some substances change colour when they react with an acid or a base. These substances are called acid-base indicators.
8. Litmus is the best known of all acid-base indicators. It does not change colour in the presence of a neutral substance, but responds to acids and bases in the following way:
 - litmus is red in the presence of an acid; and
 - litmus is blue in the presence of a base.

TOPIC 2: PROCESS TO PURIFY WATER

In this topic, we are going to explain concepts, look at the importance of drinking clean water and methods of purifying water

KEY QUESTIONS

- Why is it important to drink clean water?
- How can water be purified?

2.1 Clean water

Clean water is water that is free from any form of pollutants.

Why is it so important for humans, plants and animals to have clean water?

- Our bodies contain a lot of water. We need to drink water because we lose water every day from the waste our bodies produce. The water we drink must be clean to prevent us from becoming sick.
- Plants and animals need clean water too, so that they can grow and be healthy. We need healthy plants and animals in our environment because they provide a balance in nature and food.

2.2 Water Pollution

Water pollution is making water impure. Pollutants make water impure and are harmful to humans, animals and plants.

Examples of pollutants are: Oil and waste, chemicals and poisons, tins, dead plants and animals are all pollutants

This sign below is a warning that the water is not clean and humans should not drink it, as it could be dangerous to your health.



A warning against drinking polluted water

ACTIVITY

1. What does 'clean water' mean to you?
2. Give two examples of pollutants.

2.3 Water purification

Purify

To remove pollutants from water to make it pure.

Natural processes

Natural process is whereby undesired pollutants are removed without using chemicals e.g. as water travels through the grounds it gets filtered. Water also gets naturally purified as it flows through some kinds of an ecosystem especially in the wetlands by sediment trapping.

2.4 Different methods of purifying water

There are different methods of cleaning water: boiling, filtration, sieving, etc.

Boiling

In this method, water needs to be boiled and should be kept in a clean pot and covered. It makes it easier to kill certain microorganisms because some dies at different temperature in household system. e.g. bacteria are killed at temperature of 65°C. Boiling water to kill microorganisms.

Microorganisms are called microbes which refers to very small living things which are present in water bodies and cannot be seen by our naked eyes. Examples are: viruses and bacteria.

Filtration

This method includes removing insoluble substances from the water where the water will go through various kinds of filtering equipment.

- The water flows through a special filter made of layers of sand and gravel, only much bigger. The gravel layer of the filter is about 30 cm deep and the sand.

Sieving

- It is a method we use to separate the large impurities from the dirty water.

ACTIVITY

1. List different ways to separate mixtures.
2. Which methods do you think would be useful to separate the large impurities from the dirty water?
3. Which methods would be useful to remove the soluble impurities from the dirty water?

2.5 Which water can be purified?

Water from water bodies and not water we used from our homes.

Examples of water bodies includes: lakes, rivers, oceans and groundwater.

Imagine you are in a place without clean water. The only water around is a muddy stream. How can the dirty water be made clean enough to drink? When thinking about how to purify water, we need to consider what it is that we need to do to separate the impurities from the water. The muddy water is actually a mixture of soil and water, and possibly other impurities. Design and make your own filter to purify water.

ACTIVITY: Design, draw or make your own filter.

Practical

Water is one of the fundamental necessities of human life and as such it is important that you understand the process that goes into your everyday drinking water.

Materials

- 2-litre plastic bottle cut in half
- Napkins or paper towels
- *Gravel
- *Sand
- Cups or scoops (for gravel and sand)
- Activated Filter Carbon
- Large water container for dirty water
- Dirty water (made by adding dirt, twigs, leaves, etc., to water)
- Large waste container (plastic container or thick garbage bag for used filtering materials)
- clean jugs for filtered water

Work in groups of 4

1. Put the top half of the plastic bottle-upside down (like a funnel) inside the bottom half. The top half will be where they build their filter; the bottom half will hold their filtered water.
2. Material to be used: (gravel, sand, and carbon).

3. What will each of these layers remove?
4. Decide on an order of the filtration materials for their own filter with the exception that the napkin or paper towel should go first at the bottom (into the neck of the bottle).
5. Once you have made the filter, put the filter on a stable surface and pour in the dirty water.
6. As their filters drip water, answer the question:
What does the filtered water look like?
7. You may need to run the water through the filter a few times in order to purify the water. Excess dust from the activated filter carbon may turn the water slightly grey. In order to save time, you may want to rinse off the carbon beforehand.

ACTIVITY

4. Draw and label the water filter that you have designed indicating all the layers and uses of the different type of soil.

SUMMARY

KEY CONCEPTS

- Clean water is important for people, plants and animals.
- Water can look clean but not purified.
- Water can be purified by processes such as sieving, filtering, settling and decanting, boiling and adding chemicals to kill the microorganisms.

REVISION

1. What does it mean to purify water?
2. Why do humans, animals and plants need clean water?
3. If you were not sure about the water that came out of the tap and you had a kettle, what would you do to the water to purify it?
4. The woman in the picture is drinking water from the edge of a dam. What possible threats could she face from drinking this water without doing anything to purify it?



TOPIC 3: IMPACT OF WEATHER, TEMPERATURE AND WIND ON MATERIALS

We are going to look at the impact of weather on metals and non- metals and how to protect them against weather.

Atmospheric temperature

It is the measure of temperature at different levels of the Earth's surface atmosphere.

3.1 Impact of weather on metals and non-metals

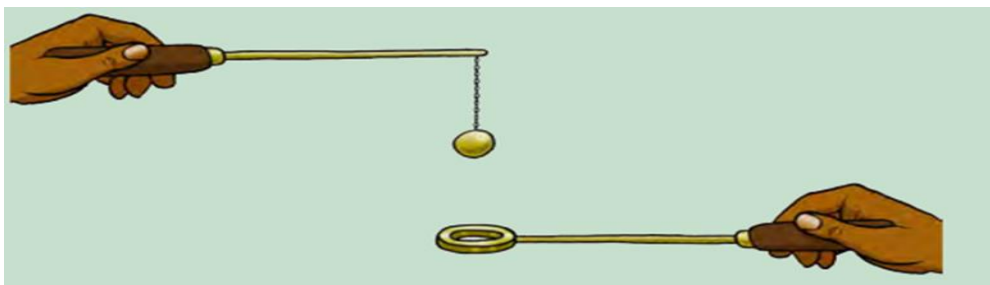
Weather

The state of the atmosphere in respect to temperature, wind, humidity and rainfall.

Weather has impact on metals through its elements.

Weather affect metals in numerous ways:

- Cold weather makes metal stiffer e.g. steel can change from ductile to brittle, instead of bending it will break.
- Humid weather harms metal more terrible than rust, this corrodes or degrade the metal, rendering it ineffective and causing self- destruction.
- Weather makes some metals rust i.e. iron.
Rust is a brown substance that forms on iron or steel, for example when it comes into contact with humidity and oxygen.
- Metal that are heated expands.
The following diagram shows a metal ball and ring apparatus. The ring and ball are both made of brass. At room temperature, the ball is just the right size to pass through the ring. Do you think the ball will still fit through the ring when it has been heated?



Let us now explore on the impact of weather on metals

3.1.1. Rusting

Rust is a brown substance that forms on iron or steel, when it comes into contact with humidity and oxygen.

How metals react in weather especially holds true for iron and steel. Even though steel is one of the most commonly used metals in the world you will often see steel rusting and that is why bridges and roadways have limited lifespan.

Weather may affect how fast or slow metal rust depending on how humid the weather is.

The diagram below shows an old car covered in rust.



An old car covered in rust

Rust has a reddish-brown colour and a rough texture.

ACTIVITY

AIM: To find out how rust occurs

MATERIALS:

- 30 identical iron nails
- 3 small clean, dry containers (yogurt tubs or polystyrene cups)
- tap water

- salt water (made by dissolving 10 teaspoons of salt in a litre of tap water)
- plastic wrap to cover the containers

METHOD:

1. Mark by writing A on the first container, B on the second one and C on the last one.
2. Place 10 iron nails in each of the containers.
3. Pour tap water on the nails in container A to cover completely.
4. Pour salt water on the nails in container B to cover them completely
5. Do not pour anything on the nails in container C.
6. Cover containers A and B with plastic wrap.
7. Place the containers next to each other in a safe spot where they can be left undisturbed for a few days.
8. Check the nails in the containers every day. Every day (preferably at the same time each day)
9. Count the number of nails that have rust on them.
10. Make sure to return the same nails to the same container after you have examined them.
11. Continue to do this over a period of 10 days.
12. Record your results in the table below

Day	Number of nails in the cup containing water only(A)	Number of nails in the cup containing salt water (B)	Number of rusty nails in the cup containing no water
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Draw a graph of how many nails had rust on them after each day.

ACTIVITY

1. In which cup did the nails start rust first?
2. Complete the following sentences. You may use the words in the box below, or any other words that will make the statement true for you.

Word box Air, water, salt water

3. Iron rusts when it comes into contact with ...
4. Iron will rust more quickly _____ than in _____
5. How can we protect iron against rust? (Hint: Look at the following picture for a clue).



These people are painting iron poles and fences

3.2. Methods to protect materials against weather.

Weather protection means temporary protection of that work adversely affected by humidity, winds, precipitation and temperature, by covering enclosing and/ or heating.

- Storing metal products at a low moisture area (inside a temperature and humidity controlled environment) will slow down rust.
- Coating iron and steel with zinc to prevents water and oxygen from reaching iron and steel.
- Painting the metal will slow down rusting by preventing moisture from reaching the metal.
- **Applying oil** to iron will help to prevent rust or slow it down, since it inhibits moisture from reaching the iron.

[illegible]

CONTENT OVERVIEW

STRAND	YEAR 2	YEAR 3	YEAR 4
Energy and Change	<u>Energy</u> - Explain concepts with examples - Different forms/types of energy	<u>Energy and Temperature</u> - Explain the concepts with examples - Describe how to read and use a thermometer - Describe how heating and cooling can change materials	<u>Electric circuits</u> - Define concepts - Explain when a circuit is complete - Conductors and Insulators
	<u>Energy Sources</u> - Explain concepts with examples - Categorise sources of energy into Renewable and Non-Renewable sources - Name the different energy sources	<u>Stored energy in fuels</u> - Explain concepts with examples - Explain how energy is stored in food - Discuss the Renewable and Non-renewable sources	<u>Systems to solve problems</u> - Identify different circuits - Make simple electrical systems
	<u>Energy Transfer</u> - Explain where the energy from food comes from - Describe how energy is transferred in a food chain - Identify different types of energy transfer	<u>Burning Fuels</u> - Investigate, compare & describe how different fuels burn: Example candle <u>Safety with fire</u> - Identify sources - Identify safety rules when using fuels	<u>Mains electricity</u> - Describe concepts with examples - Compare the amount of energy released by different energy sources
	<u>Sound and Noise pollution</u> - Explain the concept sound with different examples - Explain sound as noise pollution and the impact on the environment	<u>Energy and Electricity</u> - Briefly define concepts - List the different types of batteries and its uses	<u>Saving Cost of Generating Electricity</u> - Explain concepts with examples - Describe energy sources and usage in South Africa - Compare different electrical appliances

CONTENT OUTLINE

WEEK	TOPIC	CONTENT
1-2	Electric circuits	Electric circuits - Define concepts - Static electricity - Electric circuit - Current electricity - Electric components - Conductor - Insulator - Explain when a circuit is complete. - Demonstrate examples of conductors and insulators
3-4	Systems to solve problems	Systems to solve problems - Briefly define the following concepts and compare the strength of the current of their cells: - Series - Parallel Practical activity: Make simple electrical systems: - Draw, label or build an electric circuit that include all components - Explain and draw a diagram how electric circuits in our homes are wired from the power supplied from Eskom
5-6	Mains electricity	Mains electricity - Describe concepts with examples: - Electric supply - Generator - Turbine - Discuss with examples the different energy sources that are used to produce electricity
7-8	Saving cost of generating electricity	Saving Cost of Generating Electricity - Briefly explain concepts: - Saving - Appliance - Solar energy - Load shedding - Investigate alternative energy sources and how to use

		it instead of using electricity. Recall better practices in using energy to save cost.
9 - 10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: - Practical: 50 % - Test (Pen and Paper): 50 %		

DRAFT

TOPIC 1: ELECTRIC CIRCUITS

KEY QUESTIONS

- What is an electric circuit?
- What are the components of an electric circuit?
- What are the functions of each component?
- When is a circuit complete?
- What are conductors and insulators?

Electricity

There are two main types of electricity namely:

- Static electricity
- current electricity

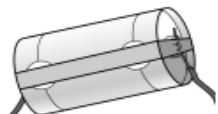
Static electricity is an imbalance of electric charges within or on the surface of a material, or between materials. When two surfaces are rubbed together there is **friction** between them. Friction is a resistance against the movement of an object because of its contact with another object.

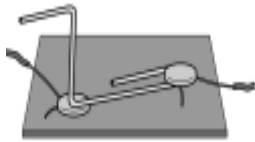
Current electricity

An electric current is the movement of charges through a conductor in a closed circuit.

An electric circuit is a closed, complete electric pathway or system for transferring electrical energy.

- Electric circuit is made up of various components as follows:

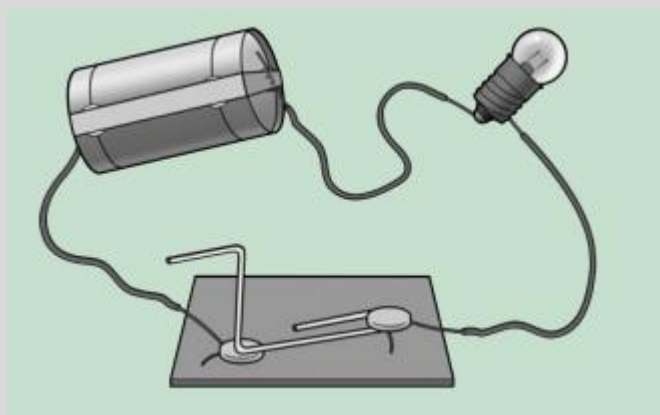
Component	Sketch	Symbol	Function
Cell			Provide energy to the circuit
Battery			Provide energy to the circuit
Bulb			Turns energy into light.
Electrical wire			Electrical wire allows the electricity to flow from one device to the next.

Switch		Open switch  Closed switch 	A Switch is used to open and close a circuit. If the switch is open, no current is flowing in a circuit. If the switch is closed, the current is flowing in a circuit.
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ACTIVITY: A simple circuit

INSTRUCTIONS

1. Look at the example of a simple circuit. And answer the questions that follow:



QUESTIONS

1. What are the components that make up this system for transferring electrical energy?
2. Do you think this is an open or closed circuit? Explain your answer.
3. Which part is providing the source of energy?
4. What is the conducting material?
5. What is the output of this system?

A circuit is a complete conducting pathway for electricity. It goes from one terminal of a cell along conducting material, through a device, and back to the other terminal of the cell.

Conductors

An electrical conductor is a material or substance that allows an electrical current to pass through it easily.

Electrical conductors are usually **metals** because metals have high conductivity. Examples are: copper, steel, iron etc.

Insulators

Insulators are non-conducting materials that do not easily allow current to pass through them. Examples of insulators are **plastic, rubber, wood etc.**

TOPIC 2: SYSTEMS TO SOLVE PROBLEMS

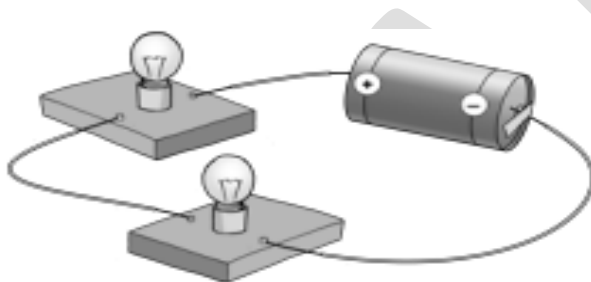
KEY QUESTIONS

- How can we use electric circuits to make useful devices and appliances work?
- What happens to energy when we use electrical devices and appliances?

Series and parallel electric circuits

Series electric circuits

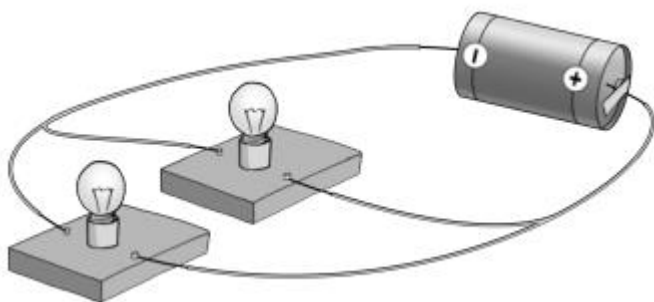
- A series circuit is one in which there is only one pathway for the electric current to follow. The components are arranged one after another in a single pathway. When we connect the components, we say that they are connected in series.



- In a series circuit, all the electrons travel through every component and wire as they travel through the circuit. All the electrons experience the same resistance and so they all move at the same rate.
- Increasing the number of cells connected in series, increases both the current strength through the circuit and the potential difference across the cells.
- The current strength in a series circuit is the same throughout the entire circuit.

Parallel electric circuits

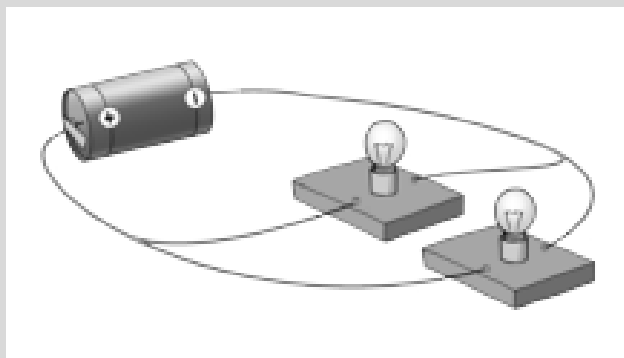
A parallel circuit provides more than one path for the electric current to move through the circuit.



- Increasing the number of cells connected in parallel with each other has no effect on the current strength and the potential difference of the circuit.
- Parallel circuits are used in the lighting systems in buildings.
-

ACTIVITY: Drawing a circuit diagram

1. Use symbols to draw a circuit diagram of the sketch above.



Using electric circuits

Electric circuits are used all around us, for example in lights in our homes, streets, and shops. Electric circuits are often used to solve a problem where we need energy. A battery or mains supply is a source of energy.

The energy is transported to a device or an appliance using electric wires. The device or appliance changes the electrical energy to another form, such as light energy, sound energy, heat energy or movement energy. We call this an energy transformation.

One form of energy is changed into another form. Appliances can do this job for us.

A system is something that consists of different parts working together to form a whole. In an electric system, there is an input (electrical energy) and an output (such as light, heat, movement, or sound energy.)

Let us look at some examples of electric systems where we use the output energy to do something useful.

Each of the examples in the Activity above uses an **electric circuit** to provide us with **energy** where we need it, whether it is light on the street, sound for our radio, or heat in a building.

We can say that electrical appliances can solve problems for us. They can transform electrical energy into another useful form of energy.

Electrical systems

Draw, label or build an electric circuit that include all components

ACTIVITY: Draw, label and build an electric circuit that include all components

Scenario

You are an electrical engineer, and you need to produce a drawing for an electrical system to solve a problem. You need to use a circuit to produce either movement, light, sound, or heat.

Design:

Now that you know a bit more about the system you want to make you need to design how you are going to make it.

Your system has the following specifications:

1. The system must make use of an electric circuit.
2. The system must produce either movement, sound, light or heat.
3. The circuit must make use of components that you have learned about, such as cells, light bulbs, switches, wires, etc.

Your system has the following constraints:

1. You must make it in class.
2. You cannot get an adult electrician to design your project for you.
3. You must produce your own design.
4. Your system must be a model.

Answer these questions:

1. What materials and tools will you need to make it? For example, which electrical components will you need?

2. Where will you place the switch? What type of switch are you going to make?

Draw and label your drawing, showing all materials and components you are going to use for your electrical system.

Make:

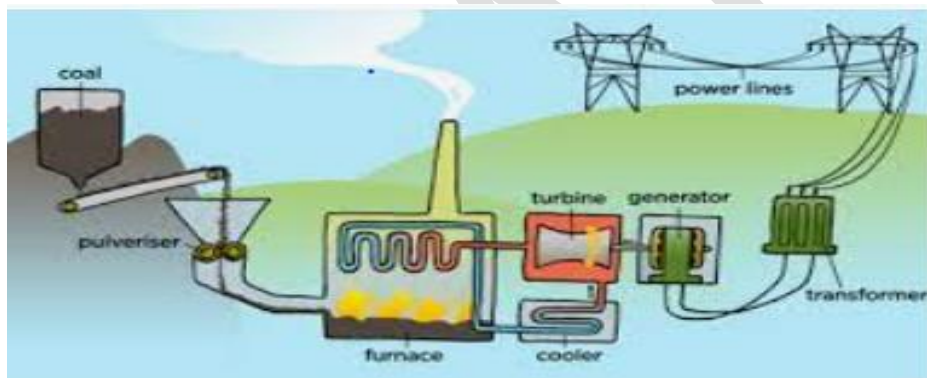
Make your system; test if it is working and present to the class.

1. Does your system look like your drawing?
2. Does your system produce a movement, sound, light or heat?
3. Where would people use the system, you made to solve a problem in their daily lives?

TOPIC 3 MAINS ELECTRICITY

Electric Supply means conductors used to transmit electric energy and their necessary supporting or containing structures.

A power supply is an electrical device that supplies electric power to an electrical load. The main purpose of a power supply is to convert electric current from a source to the correct voltage.



Fossil fuels are obtained from remains of dead plants and animals from millions of years ago. Coal, oil, and natural gas are example of fossil fuels. Fossil fuels were once alive.

Renewable Energy Sources: Renewable energy, often referred to as clean energy, comes from natural sources

Non-renewable energy: A source of energy that will eventually run out.

Most sources of non-renewable energy are fossil fuels, such as coal, gas, and oil.

WHERE DOES MAINS ELECTRICITY COME FROM?

A battery has stored energy which can provide electrical energy. However, our homes, schools, shops, and factories cannot run on batteries because they cannot store or provide large amounts of energy.

The main supplier of electrical energy is from power stations. We call this 'mains electricity'. Power stations also need a source of energy to make electricity. In South Africa, this is mostly from fossil fuels.

ACTIVITY: Different energy sources that are used to produce electricity

INSTRUCTIONS:

Read the text below about how fossil fuels were formed and study the pictures. Then answer the questions that follow.

300 million years ago...

Ancient trees, ferns and plants grew everywhere.



An ancient, prehistoric world

When these prehistoric plants and animals died their bodies decomposed just in the same way as organisms decompose today. The dead organisms became buried under layers and layers of mud, rock, sand, and water.

These layers built up and became very deep and they pushed down with a great pressure on the layers below it.

The dead plants and animals slowly decomposed and formed **fossil fuels**.

Different types of fossil fuels were formed depending on different factors depending on the temperature and pressure conditions during the decay of the organisms. They are Oil, Natural gas and Coal. The focus will be on **Coal** which is used to generate electricity.

Coal

In South Africa most of the power stations use coal as a source of energy to generate electricity. It is mined out of the earth and transported to the power station in large trucks or trains.

Coal is a black rock that can be burnt to produce energy in power stations all over the world.

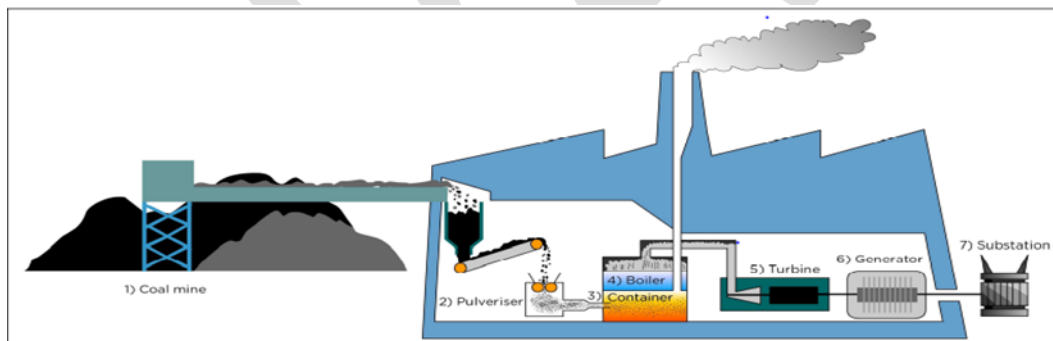
The energy in coal originally comes from energy from the Sun.

QUESTIONS:

5. Briefly explain the following:
 - a. Renewable source of energy
 - b. Non-renewable source of energy
6. Write one advantage and a disadvantage of using wind as a source of energy to generate electricity.
7. Write one advantage and a disadvantage of using Coal as a source of generating electricity in Eskom power stations.
8. What are the three fossil fuels? .
9. What are the two main factors which turned the remains of organisms into fossil fuels deep under the layers of rock and mud?
10. Explain why we say that all our energy originally comes from the sun, even in fossil fuels.

FOSSIL FUELS AND ELECTRICITY

The main supplier of electricity in South Africa is **ESKOM**. ESKOM uses mainly coal to produce energy for industrial and household use. Let us look at a power station to find out how coal is used to produce electricity. Look at the diagram and the steps which outline the process to make electricity from coal:



The process of making electricity from coal in a power station

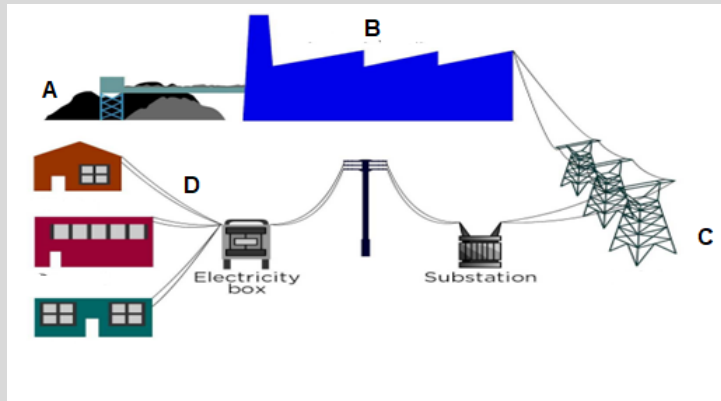
1. Coal is transported from a coal mine to a power station.
2. At the power station, the coal is ground into a fine powder (pulverised).
3. The ground coal then goes into a container where it is burned.
4. The heat generated from the burning coal is used to boil water in a huge boiler.
5. The boiling water produces steam that turns a turbine (a turbine is a big wheel which turns). **Turbine:** A turbine is a device that transforms rotational energy from a fluid that is picked up by a rotor system into usable work or energy.
6. The turbine is linked to a generator which uses a coil to produce energy. **Generator:** A generator consists of a very large solenoid with a large rotating magnet. When the magnet is turned inside the coil, the generator produces electricity. power lines. We use the energy in our homes to make our appliances work.

7. From the generator the electric current is transported ("carried") by a system of electrical transmission lines (also called power lines) and substations to our homes. The electricity is then sent to our homes through the national grid

ACTIVITY:

INSTRUCTIONS:

1. Look at the following diagram and label A to D to show how electricity is generated from power station to our homes.



TOPIC 4: SAVING COST OF GENERATING ELECTRICITY

Saving is the effort made to reduce the consumption of energy by using less of an energy service.

Appliance: a device, machine, or piece of equipment, especially an electrical one that is used in the house, such as a cooker or washing machine.

Solar energy: Solar energy is radiant light and heat from the Sun that is harnessed using a range of technologies such as solar power to generate electricity, solar thermal energy including solar water heating,

Load shedding: Load shedding is **a way to distribute demand for electrical power across multiple power sources**. Load shedding is used to relieve stress on a primary energy source when demand for electricity is greater than the primary power source can supply.

Why is electricity expensive?

Electricity is costly because:

- The production and delivery of electricity requires infrastructure (the structures and facilities) like coal mines, trucks, and trains to transport coal, power stations, substations, and wiring
- All these buildings, structures, materials, and processes are very expensive to build and maintain.

- Some electrical appliances require a lot of energy, much more than others. For example, a geyser uses a lot of electricity to heat the water and so it becomes expensive.

ACTIVITY: Conserving electricity

INSTRUCTIONS:

Look at the grid below. If the instruction helps to SAVE electricity, colour it in **BLUE**. If the statement WASTES electricity, colour it in **RED**.

Turning off appliances when on holiday	Leaving lights on in an empty room	Using an electric blanket	Using fluorescent lights
Wearing jerseys and warm clothes	Leaving outside lights on during the day	Using an electric toothbrush	Running full loads in the washing machine
Boiling a full kettle	Using a gas heater	Hanging clothes outside to dry	Turning the TV off when no one is watching
Running half loads in a dishwasher	Leaving the oven on when nothing is cooking	Using an electric can opener	Running an air conditioner with the windows open

Look at all your **RED** blocks. Rewrite each statement so that it changes from a waste of electricity to a method of saving electricity. The blocks which learners should identify as energy wasting are listed below, with a suggestion for how it can be rewritten to be energy saving:

ACTIVITY

INSTRUCTIONS:

1. Find the appliances or devices listed in the table below.
2. Copy the following table in your exercise books.
3. Record the number that is followed by a W on the label in column 2 of the table. This number indicates how much energy is required by the device in a certain time. It is called the power required by the device. The higher the value the more energy the device uses in a specific time.

Appliance or device	Power in watt (W)
Cell phone charge	36 W
Electric kettle	2 000 W
Television set	60 W
Light bulb	100 W
Energy-saving light bulb	15 W
Computer	115 W
Electric iron	1 400 W

4. Arrange the appliances in the next table in terms of the power required. The list should be from small to large values of the power.

Appliance or device	Power in watt (W)

DISCUSSION QUESTIONS:

1. Which two appliances have the lowest power requirements in the table? .
2. Which two appliances have the highest power requirements?
3. Name three (3) ways of preventing wastage of electricity in our home?

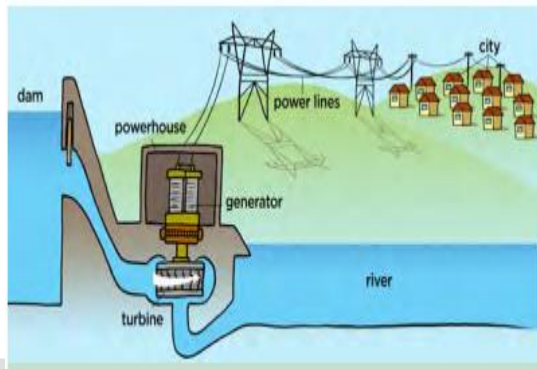
Alternative energy sources and how to use them instead of using electricity

Wind turbines convert the kinetic energy in the wind into mechanical power. This mechanical power can be used for specific tasks (such as grinding grain or pumping water) or can be converted into electricity by a generator.



Example of wind turbine

Hydropower



Example of a hydroelectric power station.



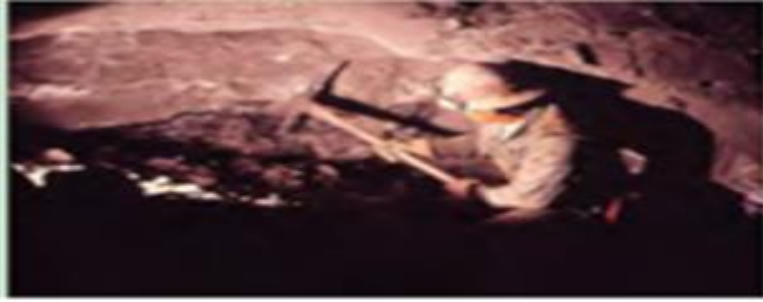
The generators inside the hydroelectric power plant.



The water then flows out the bottom of the power station and continues down the river.

Water can also be used as an energy source. This is called **hydropower**. The energy from falling water is used to drive turbines in a power station. Unlike coal power stations, the water does not need to be heated and the water can be reused. These power stations must be at waterfalls or dams because there needs to be a strong flow of water to harness the energy

Uranium



Uranium – mining for uranium underground

Uranium is the fuel most widely used to produce nuclear energy. In a nuclear reactor the uranium fuel is assembled in such a way that a controlled fission chain reaction can be achieved.

DRAFT